

High-flow injection moulding grade (Food Contact)

POKETONE Polymer M330F

POKETONE Thermoplastic Polymers are aliphatic polyketones, a revolutionary new class of semi-crystalline thermoplastics. Hyosung developed new catalyst to produce this unique polymer in 2013 and constructed commercial plant in 2015, in Ulsan, Korea.

POKETONE Polymer M330F is a high-flow injection moulding grade with mechanical properties which classify it as an engineering thermoplastic.

This grade exhibits very good processability, good impact resistance, high resilience and good creep performance. POKETONE Polymer M330F can also withstand short-term exposure to elevated temperatures.

Moreover This polymer exhibits high resistance to hydrocarbons, solvents, salt solutions, weak acids and weak bases.

POKETONE Polymer M330F is a high-flow, low-viscosity polymer that should be considered for mouldings with long flow paths or thin walls. This grade is very easy to process on standard injection moulding equipment.

Cycle times are generally short. Parts show good mould definition with glossy mar-resistant surfaces. POKETONE Polymer's low moisture sensitivity means that no conditioning of parts before assembly or use is necessary.

Applications for POKETONE Polymer M330F may be found in water related part, food contact applications, industrial and consumer appliance markets.

TABLE 1 : TYPICAL MECHANICAL PROPERTIES
OF POKETONE POLYMER M330F – Measured at 23°C

	Test Method & Conditions		ASTM Values	ISO Values
	ASTM	ISO	SI	SI
Tensile strength at yield	D638	527-1	60 MPa	60 MPa
Tensile modulus	D638	527-1	1,600 MPa	1,500 MPa
Tensile elongation at yield	D638	527-1	21%	21%
Tensile elongation at break	D638	527-1	300%	300%
Flexural strength	D790	178	57 MPa	57 MPa
Flexural modulus	D790	178	1,500 MPa	1,400 MPa
Unnotched Charpy impact strength	-	179/1eU	-	N.B.
Notched Charpy impact strength				
23°C				8 kJ/m ²
-10°C	-	179/1eA	-	4 kJ/m ²
-30°C				2 kJ/m ²
Unnotched Izod impact strength	D256	180/U	N.B.	N.B.
Notched Izod impact strength				
23°C			95 J/m	7 kJ/m ²
-10°C	D256	180/A	60 J/m	4 kJ/m ²
-30°C			40 J/m	3 kJ/m ²
Falling dart impact strength at 23°C	-	6603-2	-	50 J

TABLE 2: TYPICAL PHYSICAL PROPERTIES
OF POKETONE POLYMER M330F – Measured at 23°C

	Test Method & Conditions		ASTM Values	ISO Values
	ASTM	ISO	SI	SI
Specific gravity	D792	1183	1.24g/cm ³	1.24g/cm ³
Shore D hardness	D2240	868	-	77
Hardness Rockwell	D785	-	110	-
Water absorption equilibrium at 50% RH	D570	62	0.5%	0.5%
Water absorption at saturation	D570	62	2.1%	2.1%

TABLE 3: TYPICAL THERMAL PROPERTIES
OF POKETONE POLYMER M330F

	Test Method & Conditions		ASTM Values	ISO Values
	ASTM	ISO	SI	SI
Melting temperature	D3418	11357	222°C	222°C
Coefficient of linear thermal Expansion, 25°C to 55°C	E831	-	9.7*10 ⁻⁵	-
Vicat softening point	D1525 5 kg	306/B50 50 N	195°C	190°C
Heat deflection temperature	D648 66psi 264psi	75 0.45 MPa 1.8 MPa	200°C 105°C	190°C 92°C

TABLE 4: TYPICAL PROCESS RELATED PROPERTIES
OF POKETONE POLYMER M330F

	Test Method & Conditions		ASTM Values	ISO Values
	ASTM	ISO	SI	SI
Melt flow index 240 °C /2.16kg	D1238	1133	60 g/10 min	56mℓ/ 10min
Mould shrinkage	D955			
	MD, 3 mm		2.0%	
	TD, 3 mm	-	2.0%	-
	MD, 2 mm		1.6%	
	TD, 2 mm		1.5%	

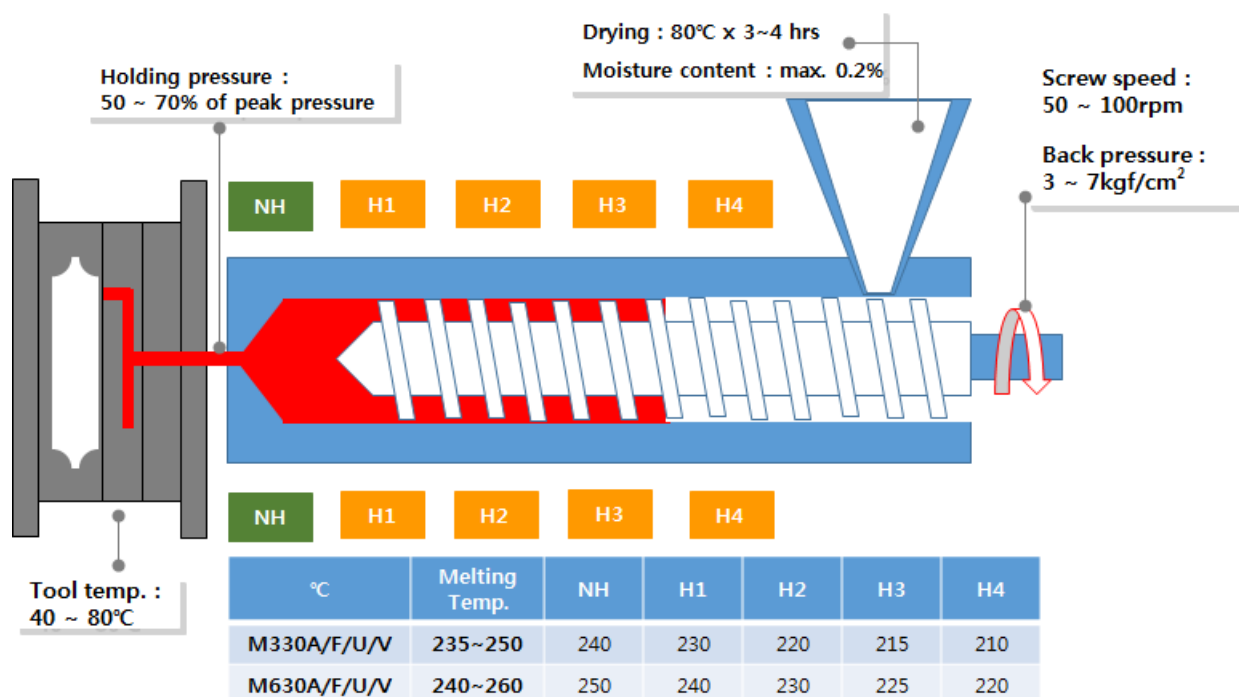
TABLE 5: TYPICAL ELECTRICAL PROPERTIES
OF POKETONE POLYMER M330F

	Test Method & Conditions		ASTM Values	IEC Values
	ASTM	IEC	SI	SI
Dielectric strength, Short term	D149 3 mm 2 mm	-	15 kV/mm 19 kV/mm	-
Volume resistivity	D257	-	10 ¹⁴ ohm cm	-
Surface resistivity	D257	-	10 ¹⁷ ohm/sq	-
Arc resistance	D495	-	130 sec	-
CTI	UL 746A	112	-	600V
Dielectric constant at 60Hz	D150	250	6.2	5
Dissipation factor at 60Hz	D150	250	0.008	0.013

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POKETONE Injection Processing Guide



Setting Temperature

- Recommended melting temperature: 235-250°C (460-490°F)
- Do not exceed 265°C (509°F). Long residence times at high end of the temperature range can cause thermal degradation & loss of physical properties.
- Mold Temperature: regarding POKETONE base grade, recommended setting temperature is at 60-80°C. In case of POKETONE glass-fiber reinforced grades, the temperature should be higher at least over 120°C for better surface quality.

Cleaning Guide

- Please immediately clean barrels thoroughly after producing POKETONE products. Recommend high viscosity HDPE, PCTG and PP (Hyosung R200P). Other commercial purging compounds are also available.

Drying

- Recommend drying POKETONE pellet at 80°C for about 3~4 hours. POKETONE should be dried by an oven or hopper drier to prevent surface problem like silver streak, drooling or voids.
- If the drying temperature is too high or the drying time is too long, it would be able to bring about discoloration of pellets.

If you need any further technical information, please contact our sales or marketing team who will be happy to assist you with any questions you may have. Feel free to visit our website. www.poly-ketone.com